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TRANSPORT PHENOMENA II, CHE 312

QUIZ 4

CLOSED BOOK, TIME: 20 MINUTES

Liquid B flows in laminar motion down a vertical wall, with the velocity profile

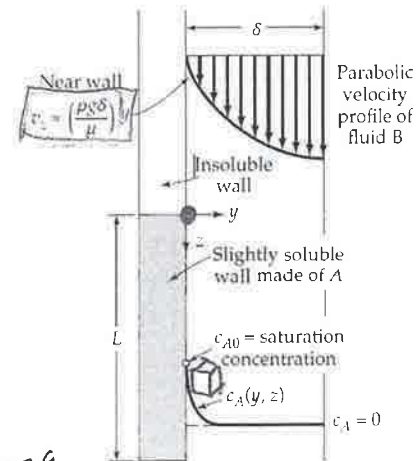
$$v_z(y) = \frac{\rho g \delta^2}{2\mu} \left[2 \left(\frac{y}{\delta} \right) - \left(\frac{y}{\delta} \right)^2 \right]$$

For $0 < z < L$ the wall is coated with a species A that is slightly soluble in B ($x_A \ll 1$). For short distances downstream, species A will not diffuse very far into the falling film. That is, A will be present only in a very thin boundary layer near the solid surface.

Write a shell mole balance for species A within the liquid film and derive a partial differential equation for the concentration of A as a function of y and z.

$L, C = \text{constant}$

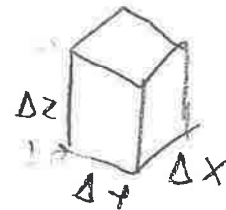
Hint: combined mole flux is $\langle N_A \rangle = c_A \langle v^* \rangle + \langle J_A^* \rangle$, where the diffusive flux according to Fick's law is $\langle J_A^* \rangle = -c D_{AB} \langle \nabla x_A \rangle$.



$$0 = \int_S \langle N_A \rangle n + 0$$

$$\checkmark \bullet N_{Ay} = -D_{AB} \frac{\partial c_A}{\partial y} + 0$$

$$\checkmark \bullet N_{Az} = c_A \langle v^* \rangle + 0 = x_A (N_A + N_B) = c x_A$$



$$0 = \int \int N_{Ay} dz dx + \int \int N_{Az} dy dz$$

$$0 = [\Delta z \Delta x N_{Ay} - \Delta z \Delta x N_{Ay+\Delta y}] + [\Delta x \Delta y N_{Az} - \Delta x \Delta y N_{Az+\Delta z}]$$

Div by $\Delta x \Delta y \Delta z$

$$0 = - \frac{\partial N_{Ay}}{\partial y} - \frac{\partial N_{Az}}{\partial z}$$

$$0 = + D_{AB} \frac{\partial^2 c_A}{\partial y^2} - \frac{\partial c_A \langle v^* \rangle}{\partial z}$$

$$\langle v^* \rangle \frac{\partial c_A}{\partial z} = D_{AB} \frac{\partial^2 c_A}{\partial y^2} \Leftrightarrow \frac{\langle v^* \rangle}{D_{AB}} \frac{\partial c_A}{\partial z} = \frac{\partial^2 c_A}{\partial y^2}$$

$$\frac{\rho g \delta^2}{\mu D_{AB}} \frac{\partial c_A}{\partial z} = \frac{\partial^2 c_A}{\partial y^2}$$